AdaBoost-Style Weighted Ensemble for Tabular Regression

Model Research Report (Sample)

1. Overview

This report describes a simple boosting-style ensemble for tabular regression. The model sequentially fits T regression trees. Each tree is trained on residuals of the current ensemble, and predictions are the sum of the initial mean prediction and a learning-rate-scaled sum of tree outputs.

2. Model Formulation

Let the dataset be $\{(x_i, y_i)\}$ with y_i a real value. Initialize $F_0(x) = \text{mean}(y)$. For $t = 1..T$ compute the pseudo-residual $r_i = y_i - F_{t-1}(x_i)$, fit a regression tree $h_t(x)$ to $\{(x_i, r_i)\}$ with maximum depth d, and update $F_t(x) = F_{t-1}(x) + \text{eta} * h_t(x)$, where eta is the learning rate. The final prediction is $F_T(x)$.

3. Hyper-parameters

Number of boosting rounds $T = 50$. Learning rate eta = 0.1. Maximum tree depth $d = 3$. The model uses squared error loss. No feature scaling is required because trees are scale invariant.

4. Training and Evaluation

Train on 80% of the rows and evaluate on the remaining 20%. Report Mean Squared Error (MSE) and R-squared on the test split. Use 5-fold cross-validation on the training split to confirm stability.

5. Implementation Notes

Implement the model in Python with scikit-learn's DecisionTreeRegressor as the base learner. Provide a class BoostingEnsemble with fit(X, y) and predict(X) methods, plus a runnable script that loads a CSV, trains, and prints MSE and R2.